NED ACADEMY

Python Practice Sheet: Lists & Tuples

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Class Level: Beginner / PGD **Focus:** Lists, Tuples, Methods & Practice

Basic List Practice

Program 1 - Create and Print a List

```
fruits = ["Apple", "Mango", "Banana", "Orange"]
print(fruits)
```

Student task: Add two more fruits and print the updated list.

Program 2 - Access List Elements

```
colors = ["Red", "Blue", "Green", "Yellow"]

print(colors[0])
print(colors[2])
print(colors[-1])
```

Student task: Predict the output of colors[-2].

Program 3 - Change a List Element

```
students = ["Ali", "Ahmed", "Sara"]
students[1] = "Hamza"
print(students)
```

Concept: Lists are mutable, so their elements can be changed.

Program 4 - Add an Item with append()

```
cities = ["Karachi", "Lahore", "Islamabad"]
cities.append("Quetta")
print(cities)
```

Student task: Add one more city using append().

Program 5 - Insert an Item

```
numbers = [10, 20, 40, 50]
numbers.insert(2, 30)
print(numbers)
```

Observe how insert() adds a value at a specific index.

Program 6 - Remove an Item

```
fruits = ["Apple", "Mango", "Banana", "Orange"]
fruits.remove("Banana")
print(fruits)
```

Student task: Remove one more fruit.

Program 7 - Sort Numbers

```
marks = [78, 92, 65, 88, 71]
marks.sort()
print(marks)

marks.sort(reverse=True)
print(marks)
```

Practice both ascending and descending sorting.

Program 8 - Find List Length

```
subjects = ["Python", "Database", "AI", "Machine Learning"]
print("Total subjects:", len(subjects))
```

Student task: Add another subject and check the new length.

Program 9 - List Slicing

```
numbers = [10, 20, 30, 40, 50, 60]
print(numbers[1:4])
print(numbers[:3])
print(numbers[3:])
```

Observe that the end index is not included in a slice.

Basic Tuple Practice

Program 10 - Create a Tuple

```
days = ("Monday", "Tuesday", "Wednesday", "Thursday", "Friday")
print(days)
```

Remember: Tuples are ordered and immutable.

Program 11 - Access Tuple Elements

```
cities = ("Karachi", "Lahore", "Islamabad", "Quetta")
print(cities[0])
print(cities[2])
print(cities[-1])
```

Student task: Print the second-last element.

Program 12 - Tuple Slicing

```
numbers = (10, 20, 30, 40, 50, 60)
print(numbers[1:4])
print(numbers[:3])
print(numbers[3:])
```

Tuple slicing works in the same way as list slicing.

Program 13 - count() in Tuple

```
grades = ("A", "B", "A", "C", "A", "B")
print(grades.count("A"))
```

Student task: Find how many students received grade B.

Program 14 - index() in Tuple

```
numbers = (10, 20, 30, 40, 50)
print(numbers.index(30))
```

Student task: Find the index of 50.

Class Practice - Try Without Looking at the Solution

1	Create a list containing 5 student names. Print the first and last student.
2	Take 3 favorite movies from the user and store them in a list.
3	Create a list of five numbers and display them in ascending and descending order.
4	Create a list of five cities. Ask the user for another city and add it using append().
5	Create the tuple (10, 20, 30, 20, 40, 20) and count how many times 20 occurs.
6	Create a tuple containing the 7 days of the week and print only the weekdays using slicing.

7	Create the list [1, 2, 3, 2, 1]. Make a copy, reverse it, and check whether the original list is a palindrome.
8	Store grades ("C", "D", "A", "A", "B", "B", "A"). Count the A grades, convert the tuple to a list, and sort it.

Suggested Class Flow

Demonstrate Programs 1-6, let students independently attempt Programs 7-14, and then use the final 8 questions as guided class or lab practice.